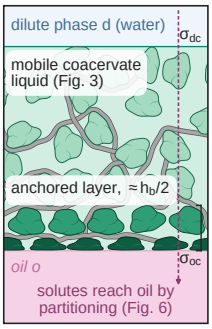
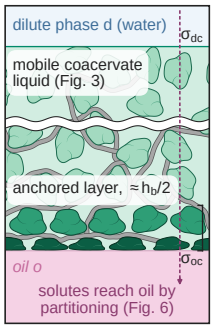
Fig. 1 molecular schematics: panels at final size (1:1, vector)

a, corona zoom (1 nm = 0.6 mm); c, film states (1 nm = 0.44 mm, all four windows). Seeds: zoom 7, film 3. Fonts embedded as text.

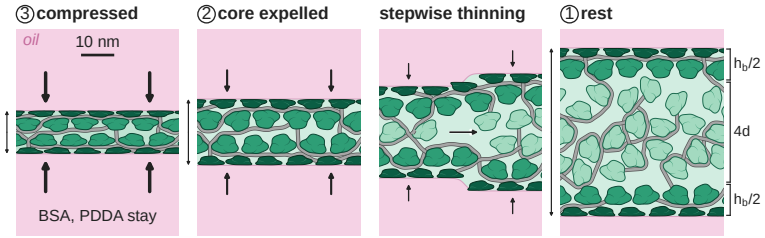
1 With labels



a zoom



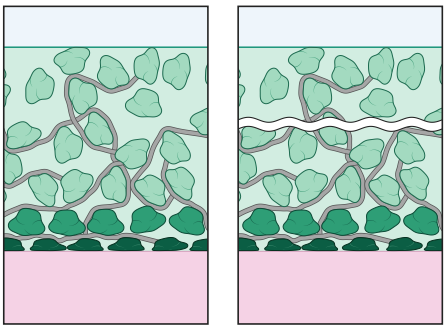
a zoom, mobile layer shortened (recommended)



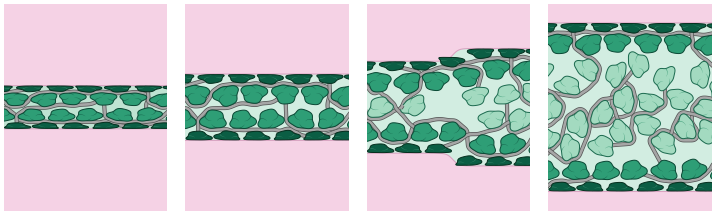
h = 13 nm (0.65 h_b) | h = h_b = 20 nm | h_b + d | h_b + 2d | h = h_b + 4d = 51 nm
Π = 0.86 MPa | Π ≈ 10 kPa | one layer out per step | Π ≈ 8–43 Pa
water, ions leave | core → Plateau borders | core = bulk coacervate

c film states, top of panel c

2 Text-free (for compositing; same molecules)



10 nm (zoom)



10 nm (film states)

3 Keys

Colour BSA adsorbed at oil · BSA anchored (glued) · BSA mobile · PDDA

Symbols
- applied pressure Π (arrow weight ~ log Π)
- film thickness h (same scale in all windows)
- step front: one mobile layer leaves
- ③②① film states marked on the Π(h) curve
- h_b/2 anchored layer; 4d mobile core
- coacervate tint; darker = less water
- break: mobile layer shortened
- solutes reach oil by partitioning (Fig. 6)

Scale 10 nm at 0.6 mm/nm (zoom) · 10 nm at 0.44 mm/nm (film states) · BSA 8 nm wide; PDDA tube 0.8 nm

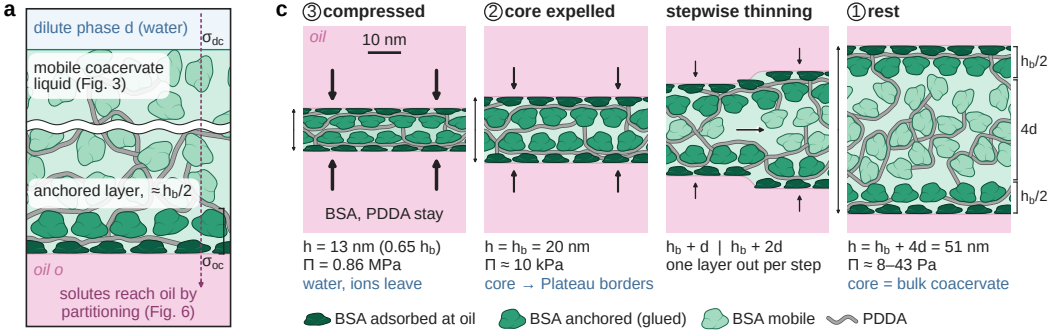
4 Draft figure-legend text for these schematics

Zoom (Fig. 1a). Molecules to scale (BSA 8 nm, PDDA 0.8 nm); in the break version the mobile layer is shortened, its thickness being set by V_C. BSA adsorbed at the oil (dark green) and a second BSA row glued onto it by PDDA (grey) form the anchored layer, ≈ h_b/2 = 10 nm. Above it the mobile coacervate is the bulk liquid: light-green BSA at area fraction ≈ 0.4 in a PDDA network of isotropic orientation. One PDDA chain (Mw 1.0–1.5 MDa, 6,200–9,300 charges) charge-matches ≈ 340–710 BSA, about 7–9 nm of chain per BSA, so each chain wraps and bridges the proteins it glues.

Film states (Fig. 1c, top). States at the points marked on Π(h), drawn at one scale and ordered by h as on the axis below. (1) Rest, h = h_b + 4d: two anchored layers and a mobile core of bulk coacervate. Between (1) and (2) the core leaves one layer at a time: a step front between h_b + d and h_b + 2d moves toward the thicker film. (2) h = h_b: the four anchored rows (same molecules as in (1)) meet and PDDA stitches across the midplane; a chain neutralises all four layers over ≈ 80–115 nm and crosses the film tens of times. (3) h = 0.65 h_b at 0.86 MPa: the same molecules and chains as (2), closer and flatter; water and ions leave, BSA and PDDA stay. Arrows in the oil, applied pressure (weight ~ log Π).

Fig. 1 molecular schematics: a, corona zoom; c, film states (Π(h) order)

Built 1:1. Zoom 1 nm = 0.6 mm; film states 1 nm = 0.44 mm (all four windows). BSA 8 nm wide; PDDA tube 0.8 nm. Green shade = mobility.



PDDA, Mw 1.0–1.5 MDa: how many BSA one chain glues

Repeat unit 161.7 g/mol, 0.50 nm, +1 → 6,185–9,278 charges; contour 3.1–4.6 µm
Coil (lp = 3 nm, ideal in the dense phase): Rg = 56–68 nm
Charge-matched (BSA net −13 to −18): 344–714 BSA per chain (≈ 499 at 1.25 MDa)
= 6.5–9.0 nm PDDA per BSA ≈ one BSA diameter: each glued BSA is wrapped
Implied PDDA 8–11 mg/mL; mesh 6.9–8.1 nm (SANS ξ ≈ 9, d = 7.7 nm)
Bulk: one coil volume holds 1,698–3,120 BSA → 4–5 chains interpenetrate

The four tight layers at hb = 20 nm

7.7-nm pitch → 1.9 mg/m² per layer (≈ a protein monolayer); 5 nm per layer
One chain neutralises all 4 layers over 81–116 nm ≈ its coil width (91–111 nm)
It crosses the 20-nm film ~46–70 times: the anchored layers are stitched into one sheet
Dry 5.7 nm; φdry 0.20 (bulk) → 0.28–0.29 (hb) → 0.44–0.53 (0.86 MPa)
→ below hb only water and ions can leave: the film densifies

Drawing rules

PDDA routed BSA to BSA: wraps each one it glues, then a tangent bridge that threads between two BSA or spans the gap (lp ≈ 3 nm)
No direction bias in the routing → isotropic where BSA are random
No chain ends in view: they leave at a window edge or go under a BSA
Adsorbed row flattened on the oil; glued row in its hollows, +1.1 nm
Same anchored molecules in every state; (3) = (2) squeezed to 0.65

What was drawn

PDDA per BSA in view: 9.2 nm (zoom), 8.9 (rest), 8.8 (hb)
Order S = <cos 2θ>: mobile +0.06 (zoom), -0.05 (rest) → isotropic anchored +0.59, film at hb +0.45 → parallel to the oil (confined)
Mobile BSA area fraction 0.39 (zoom), 0.43 (rest); φeff 0.41–0.47
5 PDDA crossings of the midplane at hb (49-nm window)